

NAME: PAVITHRA B

SRN: PES1UG25AM807

Section:C

TASK-1:**1.A**

```
rm -rf .pes
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make all
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function 'write_tree_level':
tree.c:117:13: warning: '_builtin___strncpy_chk' output may be truncated copying 255 bytes from a string of length 255 [-Wstringop-truncation]
   117 |         strncpy(te->name, dir_name, sizeof(te->name) - 1);
       |         ^
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c: In function 'index_add':
index.c:178:9: warning: implicit declaration of function 'object_write' [-Wimplicit-function-declaration]
   178 |         if (object_write(OBJ_BLOB, data, nread, &blob_id) != 0) { free(data); return -1; }
       |         ^
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
/usr/bin/ld: tree.o: in function 'tree_from_index':
tree.c:(.text+0x727): undefined reference to 'index_load'
collect2: error: ld returned 1 exit status
make: *** [Makefile:22: test_tree] Error 1
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ rm -rf .pes
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make test_objects
make: 'test_objects' is up to date.
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check
All Phase 1 tests passed.
```

1.B

```
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c: In function 'index_add':
index.c:178:9: warning: implicit declaration of function 'object_write' [-Wimplicit-function-declaration]
   178 |         if (object_write(OBJ_BLOB, data, nread, &blob_id) != 0) { free(data); return -1; }
       |         ^
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
/usr/bin/ld: tree.o: in function 'tree_from_index':
tree.c:(.text+0x727): undefined reference to 'index_load'
collect2: error: ld returned 1 exit status
make: *** [Makefile:22: test_tree] Error 1
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ rm -rf .pes
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make test_objects
make: 'test_objects' is up to date.
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check
All Phase 1 tests passed.
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

TASK-2:**2.A**

```

int tree_parse(const void *data, size_t len, Tree *tree_out) {
    tree_out->count = 0;
    const uint8_t *ptr = (const uint8_t *)data;
    EOF return write_tree_level(index.entries, index.count, "", id_out);return
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make all
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function 'write_tree_level':
tree.c:126:13: warning: '_builtin___strncpy_chk' output may be truncated copying 255 bytes from a string of length 255 [-Wstringop-truncation]
   126 |         strncpy(te->name, dir_name, sizeof(te->name) - 1);
       |         ^
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c: In function 'index_add':
index.c:178:9: warning: implicit declaration of function 'object_write' [-Wimplicit-function-declaration]
   178 |         if (object_write(OBJ_BLOB, data, nread, &blob_id) != 0) { free(data); return -1; }
       |         ^
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make commit.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ rm -rf .pes
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make test_tree
make: 'test_tree' is up to date.
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.

```

2.B

```

pavii@pavithra:~/PES1UG25AM807-pes-vcs$ find .pes/objects -type f
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ rm -rf .pes
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ echo "hello" > file1.txt
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ ./pes add file1.txt
Segmentation fault (core dumped)
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ ./pes commit -m "test"
[main 18b0b0f] test
Committed: 18b0b0f97f9a... test
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ find .pes/objects -type f
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/18/b0b0f97f9ae89a28a2caaaa3294bae570cabec02a5390af1c0306ff5821f17
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ xxd .pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321 | head -20
00000000: 7472 6565 2030 00                                tree 0.

```

TASK-3:

3.A

```

untracked: engine.c
untracked: commit.h
untracked: tree.h
untracked: file2.txt
untracked: pes.c

pavii@pavithra:~/PES1UG25AM807-pes-vcs$ rm -rf .pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:   file1.txt
  staged:   file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
untracked: README.md
untracked: .gitignore
untracked: test_sequence.sh
untracked: Makefile
untracked: test_objects
untracked: test_objects.c
untracked: index.c
untracked: test_tree
untracked: object.c
untracked: index.h
untracked: tree.c
untracked: pes.h
untracked: commit.c
untracked: test_tree.c
untracked: engine.c
untracked: commit.h
untracked: tree.h
untracked: pes.c

```

3.B

```
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ rm -rf .pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
  untracked: README.md
  untracked: .gitignore
  untracked: test_sequence.sh
  untracked: Makefile
  untracked: test_objects
  untracked: test_objects.c
  untracked: index.c
  untracked: test_tree
  untracked: object.c
  untracked: index.h
  untracked: tree.c
  untracked: pes.h
  untracked: commit.c
  untracked: test_tree.c
  untracked: engine.c
  untracked: commit.h
  untracked: tree.h
  untracked: pes.c

pavii@pavithra:~/PES1UG25AM807-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776848247 6 file1.txt
100644 e08c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776848247 6 file2.txt
```

TASK-4:

4.A

```
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776848247 6 file1.txt
100644 e08c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776848247 6 file2.txt
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ rm -rf .pes
./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"
echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"
echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
./pes log
Initialized empty PES repository in .pes/
[main dd8f8d8] Initial commit
Committed: dd8f8d831930... Initial commit
[main 5171d04] Add world
Committed: 5171d04195f6... Add world
[main 6fa2324] Add farewell
Committed: 6fa23244a596... Add farewell
commit 6fa23244a596529d9331a5a7dc4a4a54a1543fcd81c6303877370856653e7be2
Author: Your Name <PES1UG25AM807>
Date: 1776848313

    Add farewell

commit 5171d04195f64a0dbcf77b5b7d0bad553c6cc7aebc07407fb0aeb0c875ff29f22
Author: Your Name <PES1UG25AM807>
Date: 1776848313

    Add world

commit dd8f8d8319306ae2d6636aa16c533682465d40452873b17633ecf4626d2553f0
Author: Your Name <PES1UG25AM807>
Date: 1776848312

    Initial commit
```

4.B

```
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/51/71d04195f64a0dbcf77b5b7d0bad553c6cc7aebc07407fb0aeb0c875ff29f22
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6f/a23244a596529d9331a5a7dc4a4a54a1543fcd81c6303877370856653e7be2
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/dd/8f8d8319306ae2d6636aa16c533682465d40452873b17633ecf4626d2553f0
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4.C

```
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/51/71d04195f64a0dbc77b5b7d0bad553c6cc7aebc07407fb0aeb0c875ff29f22
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6f/a23244a596529d9331a5a7dc4a4a54a1543fcd81c6303877370856653e7be2
.pes/objects/a6/882ee4afd855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/dd/8f8d8319306ae2d6636aa16c533682465d40452873b17633ecf4626d2553f0
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
6fa23244a596529d9331a5a7dc4a4a54a1543fcd81c6303877370856653e7be2
ref: refs/heads/main
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ cat .pes/refs/heads/main
6fa23244a596529d9331a5a7dc4a4a54a1543fcd81c6303877370856653e7be2
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

4.D

```
pavii@pavithra:~/PES1UG25AM807-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:    file.txt
  staged:    hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)

--- First Commit ---
[main fe550fe] Initial commit
Committed: fe550fe502bb... Initial commit

Log after first commit:
commit fe550fe502bb1882f8fd5bdb40c3cc4b289d9dd40e206ce78228eb5c1946c545
Author: Your Name <PES1UG24AM129>
Date:   1776848468

    Initial commit

--- Second Commit ---
[main fc91860] Update file.txt
Committed: fc918605aff7... Update file.txt

--- Third Commit ---
```

```

Initial commit

--- Second Commit ---
[main fc91860] Update file.txt
Committed: fc918605aff7... Update file.txt

--- Third Commit ---
[main 77706af] Add farewell
Committed: 77706afb760c... Add farewell

--- Full History ---
commit 77706afb760cc0c299682f03c410a7dd6454e618905d8e0c0af9ed915582a916
Author: Your Name <PES1UG25AM807>
Date: 1776848469

    Add farewell

commit fc918605aff7adb03fdc0751eedb77d690fc652d4c0ae31d1f8523f939530439
Author: Your Name <PES1UG25AM807>
Date: 1776848469

    Update file.txt

commit fe550fe502bb1882f8fd5bdb40c3cc4b289d9dd40e206ce78228eb5c1946c545
Author: Your Name <PES1UG25AM807>
Date: 1776848468

    Initial commit

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
77706afb760cc0c299682f03c410a7dd6454e618905d8e0c0af9ed915582a916

--- Object Store ---
Objects created:
10

```

```

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/77/706afb760cc0c299682f03c410a7dd6454e618905d8e0c0af9ed915582a916
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafaaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/fc/918605aff7adb03fdc0751eedb77d690fc652d4c0ae31d1f8523f939530439
.pes/objects/fe/550fe502bb1882f8fd5bdb40c3cc4b289d9dd40e206ce78228eb5c1946c545

=== All integration tests completed ===

```